

Supplemental Online Content

Buwaneswarran S, Wong YL, Liang S, Quek SC, Lee J. Active treatment vs expectant management of patent ductus arteriosus in preterm infants: a meta-analysis. *JAMA Pediatrics*. Published online May 27, 2025. doi:10.1001/jamapediatrics.2025.1025

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eReferences

This supplemental material has been provided by the authors to give readers additional information about their work.

eTable 1. Search Strategy

PUBMED	
No.	Query
6	(#1 OR #2) AND #3 AND #5
5	"Acetaminophen"[Mesh] OR acetaminophen OR paracetamol OR "Ibuprofen"[Mesh] OR ibuprofen OR brufen OR "Indomethacin"[Mesh] OR indomethacin OR indocid OR "Cardiac Catheterization/methods"[Mesh] OR "Vascular Closure Devices"[Mesh] OR Device closure OR closure OR transcatheter OR Percutaneous OR device OR coil OR catheter OR occlusion OR "Cardiac Surgical Procedures"[Mesh] OR Surgical Ligation OR closure OR ligation OR surgical OR surgery OR clip
3	"Ductus Arteriosus"[MeSH Terms] OR "ductus arteriosus, patent"[MeSH Terms] OR "PDA"[All Fields] OR "ductus"[All Fields] OR "ductal"[All Fields] OR "Ductus Arteriosus"[All Fields] OR "arterial duct"[All Fields]
2	"Infant, Very Low Birth Weight"[Mesh] OR very low birth weight OR very-low-birth-weight
1	"Infant, Extremely Premature"[Mesh] OR Infant, extremely premature OR extremely preterm OR preterm OR infant, newborn OR infan* OR neonat*
Filters: RCT	

EMBASE	
No.	Query
#11	#8 AND #9 AND [randomized controlled trial]/lim AND [humans]/lim AND [english]/lim AND ([embase]/lim OR [preprint]/lim)
#10	#8 AND #9
#9	#5 OR #6
#8	#3 AND #4 AND #7
#7	#1 OR #2
#6	lung dysplasia'/exp OR 'bronchopulmonary dysplasia' OR 'chronic lung disease of prematurity'/exp OR 'chronic lung disease of prematurity'
#5	death'/exp OR 'death'
#4	paracetamol'/exp OR 'paracetamol' OR 'ibuprofen'/exp OR 'ibuprofen' OR 'indomethacin'/exp OR 'indomethacin' OR 'ligation'/exp OR 'ligation' OR 'clip'/exp OR 'clip' OR 'coil'/exp OR 'coil' OR 'device closure'/exp OR 'device closure' OR 'device'/exp OR 'device'
#3	patent ductus arteriosus'/exp OR 'patent ductus arteriosus' OR 'ductus'
#2	very low birth weight'/exp OR 'very low birth weight'
#1	extremely premature birth'/exp OR 'extremely premature birth'

COCHRANE	
No.	Query
#1	Infant, extremely premature OR extremely preterm OR preterm OR infant, newborn OR infan* OR neonat*
#2	very low birth weight OR very-low-birth-weight
#3	patent ductus arteriosus OR PDA OR ductus OR ductal OR ductus arteriosus OR arterial duct
#4	acetaminophen OR paracetamol OR ibuprofen OR brufen OR indomethacin OR indocid OR Device closure OR closure OR transcatheter OR Percutaneous OR device OR coil OR catheter OR occlusion OR Surgical Ligation OR closure OR ligation OR surgical OR surgery OR clip
#5	(#1 OR #2) AND #3 AND #4
#6	bronchopulmonary dysplasia OR BPD OR chronic lung disease
#7	death OR mortality
#8	#5 AND (#6 OR #7) in Trials

eTable 2. Excluded Full-Text Articles (n = 24) and Reasons for Their Exclusion

Study Author, Year	Reason for Exclusion
Härkin et al., 2016 ¹	No expectant management
Bussmann et al., 2021 ²	Post-hoc analysis of previously reported RCT
Yanagi et al., 1981 ³	No expectant management
Schindler et al., 2021 ⁴	No expectant management
Kluckow et al., 2019 ⁵	Post-hoc analysis of previously reported RCT
Ment et al., 1994 ⁶	No expectant management
Gournay et al., 2004 ⁷	No expectant management
Rudd et al., 1983 ⁸	No expectant management
Schmidt et al., 2001 ⁹	No expectant management
de Carolis et al., 2000 ¹⁰	No expectant management
Mahony et al., 1985 ¹¹	No expectant management
Hammerman et al., 2017 ¹²	No expectant management
Supapannachart et al., 1999 ¹³	No expectant management
van Overmeire et al., 2004 ¹⁴	No expectant management
Aranda et al., 2009 ¹⁵	No expectant management
Cassady et al., 1989 ¹⁶	No expectant management
Alfaleh et al., 2008 ¹⁷	No expectant management
Vincer et al., 1987 ¹⁸	No expectant management
Juujärvi et al., 2019 ¹⁹	Post-hoc analysis of previously reported RCT
Bandstra et al., 1988 ²⁰	No expectant management
Lai et al., 1990 ²¹	No expectant management
Couser et al., 1996 ²²	No expectant management
Couser et al., 2020 ²³	No expectant management
Schmidt et al., 2006 ²⁴	No expectant management

eTable 3. Individual Article Characteristics

Study, Year	Location (Number of centres)	Study Design	Inclusion	Total Sample Size, n (Study vs. Control)	Active Treatment	Open label medical treatment, n (%) (Study vs. Control)	Surgical ligation, n (%) (Study vs. Control)	Primary Outcome
Sosenko et al., 2012 ²⁵	America (1)	Double-blinded, placebo-controlled RCT	GA 23 - 32 weeks BW 500 – 1250g 24HOL – 14 days old Clinical symptoms of hs-PDA with 2DE showing predominant L-R or bidirectional shunt	105 (54 vs 51)	Intravenous Ibuprofen	14 (25.9) vs 23 (45.1)	8 (14.8) vs 10 (19.6)	Number of days of O2 during first 28 days Composite of death by 36wPMA or O2 use at 36wPMA Death by 36wPMA O2 use at 36wPMA Number of days on MV during first 28 days Need for O2 / MV at D28
Kluckow et al., 2014 ²⁶	Australia (3)	Double-blinded, placebo-controlled RCT	GA < 29 weeks 2DE before 12HOL showing PDA size > 50% of PNA reference	92 (44 vs 48)	Intravenous Indomethacin	9 (20.5) vs 19 (39.6)	0 (0) vs 2 (4.2)	Composite of death, ≥ Grade II PIVH and cystic PVL by discharge
Clyman 2019 ²⁷	America United Kingdom Sweden Turkey (17)	Placebo-controlled RCT	GA 23-25 weeks and 6-14 DOL GA 26-27 weeks and 8-14 DOL Respiratory support Moderate-large PDA by echocardiographic criteria	202 (104 vs 98)	Intravenous Indomethacin or Ibuprofen or Paracetamol	19 (18.3) vs 47 (48.0)	12 (11.5) vs 12 (12.2)	Need for surgical ligation or PDA follow-up with cardiology after discharge
Hundscheid et al., 2023 ²⁸	Netherlands Belgium Denmark (17)	Non-inferiority RCT	GA < 29 weeks < 72HOL PDA > 1.5mm with L to R predominant shunt	273 (137 vs 136)	Intravenous Ibuprofen	52 (38.0) vs 35 (25.7)	3 (2.2) vs 0 (0)	Composite of NEC IIA and above + moderate-severe BPD + death by 36wPMA

Gupta et al., 2024 ²⁹	United Kingdom (32)	Double-blinded, placebo-controlled RCT	GA 23 – 28 weeks < 72HOL PDA > 1.5mm with unrestricted L-R flow, no clinical pulmonary hypertension Clinical equipoise	653 (326 vs 327)	Intravenous Ibuprofen	43 (13.2) vs 82 (25.1)	9 (2.8) vs 31 (9.5)	Composite of death by 36wPMA or mod-severe BPD at 36wPMA
Rozé et al., 2021 ³⁰	France (11)	Double-blinded, placebo-controlled RCT	GA 24 – 27 weeks 6-12HOL Large PDA calculated using PNA criteria	228 (114 vs 114)	Intravenous Ibuprofen	20 (17.5) vs 71 (62.3)	8 (7.0) vs 15 (1.2)	Survival without CP at PMA 24mth
El-Khuffash et al., 2021 ³¹	Ireland (1)	Double-blinded, placebo-controlled RCT	GA < 29 weeks 36-48HOL Echocardiographic PDA risk score ≥ 5.0	60 (30 vs 30)	Intravenous Ibuprofen	5 (16.7) vs 11 (36.7)	6 (20.0) vs 6 (20.0)	Composite of BPD at 36wPMA and death at discharge
de Waal et al., 2021 ³²	Australia (2)	Double-blinded placebo-controlled pilot RCT	GA < 29 weeks < 72HOL PDA > 1.5mm	72 (35 vs 37)	Intravenous Ibuprofen or Indomethacin	1 (2.9) vs 1 (2.7)	0 (0) vs 0 (0)	Recruiting Rate, incidence of open-label treatment
Potsiurko et al., 2024 ³³	Ukraine (1)	Non-inferiority RCT	GA < 32 weeks BW < 1500g < 72HOL PDA > 1.5mm on 2DE	208 (104 vs 104)	Rectal Ibuprofen or Intravenous Paracetamol	0 (0) vs 8 (7.7)	0 (0) vs 0 (0)	Composite of death by 36wPMA or BPD at 36wPMA
Sung et al., 2020 ³⁴	South Korea (1)	Double-blinded, placebo-controlled RCT	GA 23 – 31 weeks Requiring respiratory support PDA > 1.5mm with predominant L-R shunt at PNA 6 – 14 days	142 (70 vs 72)	Oral ibuprofen	1 (1.4) vs 0 (0)	1 (1.4) vs 0 (0)	Composite of death by 36wPMA or BPD at 36wPMA

Abbreviations: BPD, bronchopulmonary dysplasia; BW, body weight; CP, cerebral palsy; DOL, day of life; GA, gestational age; hs-PDA, hemodynamically significant patent ductus arteriosus; HOL, hours of life; L, left; PDA, patent ductus arteriosus; PIVH, periventricular/ intraventricular hemorrhage; PMA, postmenstrual age; PNA, postnatal age; PVL, periventricular leukomalacia; NEC, necrotizing enterocolitis; O2, oxygen; MV, mechanical ventilation; R, right; RCT, randomized controlled trial; 2DE, 2D Echo.

eTable 4. Individual Articles' Secondary Outcomes and Bronchopulmonary Dysplasia (BPD) Definition

Study, Year	Secondary Outcomes	BPD definition
Sosenko et al., 2012 ²⁵	• Total duration of oxygen supplementation • Duration of mechanical ventilation • Need for oxygen supplementation >30% at 36 weeks • Pneumothorax • Pulmonary interstitial emphysema • Postnatal steroids • Intestinal perforation • NEC requiring surgery • Intracranial hemorrhage • PVL • ROP • Sepsis (defined as positive blood culture results)	Oxygen requirement at 36 weeks' postmenstrual age [Jobe & Bancalari, 2001] ³⁵
Kluckow et al., 2014 ²⁶	• Mortality • Periventricular-IVH • PVL • Total doses and side effects of indomethacin • Renal impairment (elevated creatinine >150 mmol/L, oliguria <1.0 mL/kg/h for >12 h during treatment) • GI bleeding • Spontaneous intestinal perforation • NEC – greater than stage 1 of Bell's classification • Time to full oral feeds (150 mL/kg/day) • PH (frank blood-stained respiratory secretions, a sudden clinical increase in respiratory support and chest x-ray changes) • ROP • Sepsis (positive blood culture and antibiotics >5 days) • Duration of mechanical ventilation, continuous positive airway pressure, or oxygen therapy • Duration of hospitalization • BPD (physiological requirement for oxygen at 36 weeks' GA) • Diuretics or postnatal steroid for BPD • Developmental outcome at 12 and 36 months (planned)	Physiological requirement for oxygen at 36 weeks' GA [Jobe & Bancalari, 2001; Walsh et al., 2004] ^{35,36}
Clyman 2019 ²⁷	• Duration of intubation and respiratory support • Time until achievement of full enteral intake	Modified room air challenge test between 36 0/7 and 36 6/7 weeks post-menstrual age

	• Duration of gavage feeding • Incidence of serious neonatal morbidities (NEC, BPD, death, and BPD/death)	[Jobe & Bancalari, 2001; Walsh et al., 2004] ^{35,36}
Hundscheid et al., 2023 ²⁸	• Surgical PDA ligation • Death at 28 days • PH • Pulmonary air leakage • Cardiovascular support • Renal failure • IVH • Sepsis • ROP treatment • Co-interventions (steroids, paracetamol, diuretics) • Other support (oxygen, invasive/non-invasive respiratory support) • Time till full enteral feeds	Supplemental oxygen or positive-pressure ventilatory support at 36 weeks postmenstrual age after at least 28 cumulative days of supplemental oxygen, with oxygen reduction test (or expert opinion if not performed) [Jobe & Bancalari, 2001; Walsh et al., 2004] ^{35,36}
Gupta et al., 2024 ²⁹	• Severity of BPD • Severe IVH • Cystic PVL • ROP requiring treatment • Clinically significant PH • Acute pulmonary hypertension • Definitive NEC • Closed or clinically non-significant PDA less than 1.5 mm in diameter with restricted flow at 3 weeks' age • Open-label treatment of a PDA causing symptoms • Weight gain • Discharge home while receiving supplemental oxygen	Supplemental oxygen or positive-pressure ventilatory support at 36 weeks postmenstrual age after at least 28 days of oxygen and/or respiratory support, with oxygen reduction test [Jobe & Bancalari, 2001; Walsh et al., 2004] ^{35,36}
Rozé et al., 2021 ³⁰	• Ductal status on days 3 and 14 • Open-label rescue treatment • Surgical ligation • Severe morbidities • Death and survival without severe morbidity at 36 weeks corrected age or discharge, whichever came first • Incidences and duration of mechanical ventilation, non-invasive ventilation and O2 delivery • Ages and Stages Questionnaire (ASQ) total score at 24 months' corrected age	Supplemental oxygen at 36 weeks postmenstrual age, using Walsh's room air challenge test [Jobe & Bancalari, 2001; Walsh et al., 2004] ^{35,36}

El-Khuffash et al., 2021 ³¹	• PH • Surgical PDA ligation • Culture positive sepsis • IVH status at 36 weeks corrected age • PVL • NEC • ROP requiring laser therapy • Days spent on invasive ventilation, continuous positive airway pressure, high-flow nasal cannula, and/or oxygen therapy	Oxygen supplementation at 36 weeks corrected age [Shennan et al., 1988] ³⁷
de Waal et al., 2021 ³²	• Death or chronic lung disease (CLD), defined as oxygen dependency, determined with an oxygen challenge test at 36 weeks post-conceptual age • Rate of PDA closure, defined as a PDA with a diameter <1.0 mm or closed at 10-14 days post-birth • PDA ligation • PH • IVH • NEC (≥ stage 2b) • GI bleeding • Spontaneous intestinal perforation • Duration of mechanical ventilation • Nasal continuous positive airway pressure • Hospitalization, diuretics, or postnatal steroid use for CLD • ROP necessitating treatment	Oxygen dependency, determined with an oxygen challenge test at 36 weeks post-conceptual age [Jobe & Bancalari, 2001; Walsh et al., 2004] ^{35,36}
Potsiurko et al., 2024 ³³	• Mortality • Incidence of BPD • Incidence of NEC (stage Bell's IIA) • PVL • IVH (grade ≥III) • Oliguria (<1 mL/kg/h) • Hypotension (defined as mean blood pressure < gestational age in completed weeks) • Failure to close PDA within a week after the first and second course of pharmacological treatment • Reopening rate of PDA • Need for surgical ligation of PDA • Duration of mechanical ventilation • Duration of noninvasive respiratory support • Duration of oxygen supplementation	Oxygen or respiratory support at 36 weeks postmenstrual age and/or hospital discharge [Jobe & Bancalari, 2001; Walsh et al., 2004] ^{35,36}

	• Age at administration of full volume of enteral nutrition (160 mL/kg/day)	
Sung et al., 2020 ³⁴	• BPD incidence • Death before discharge • Severe IVH (grade $\geq$ III) • ROP (stage $\geq$ 3) • NEC (stage $\geq$ IIb) • GI surgery • Nosocomial sepsis confirmed on blood culture results	Need for supplemental oxygen and/or positive pressure to maintain oxygen saturation greater than 90% at 36 weeks' GA [Reference not provided]

Abbreviations: GA, gestational age; GI, gastrointestinal; IVH, intraventricular hemorrhage; NEC necrotizing enterocolitis; PDA, patent ductus arteriosus; PH, pulmonary hemorrhage; PVL, periventricular leukomalacia; ROP, retinopathy of prematurity.

eTable 5. Differences Between A Priori PROSPERO Protocol and Final Study

<u>Included additional primary outcome (“post-hoc outcomes”)</u> 1. Death at 36 weeks or at discharge (whichever occurred later). Reason for inclusion: Not all included papers reported death at 36 weeks. Some reported death at discharge. Death during hospitalization is a clinically important outcome. Inclusion of this outcome resulted in the following additional primary outcomes: 1) Death at 36 weeks or at discharge (whichever occurred later) 2) Composite outcome of death at 36 weeks or at discharge (whichever occurred later) or moderate-severe BPD 2. Composite outcome of death at 36 weeks or moderate-severe BPD. Reason for inclusion: Clinically important outcome to report. <u>Additional statistical analysis</u> 1. Included risk difference, in addition to <i>a priori</i> plan for risk ratio, in statistical analysis of each primary and secondary outcome.
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Abbreviations: BPD, bronchopulmonary dysplasia.

eTable 6. Baseline Characteristics

Characteristic	Active Treatment (n=1018)	Expectant Management (n=1017)
Male	508/1018 (49.9%)	551/1017 (54.2%)
Gestational Age, weeks*	26.2 ± 1.7	26.3 ± 1.7
Birth Weight, grams*	874.7 ± 222.1	897.7 ± 216.5
Received surfactant	377/464 (81.2%)	374/456 (82.0%)
Multiple births	238/825 (28.8%)	248/825 (30.0%)
Antenatal Steroids	883/1018 (86.7%)	873/1017 (85.8%)
Open label medical treatment	164/1018 (16.1%)	297/1017 (29.2%)

Data are presented as mean±SD or n/n (%).

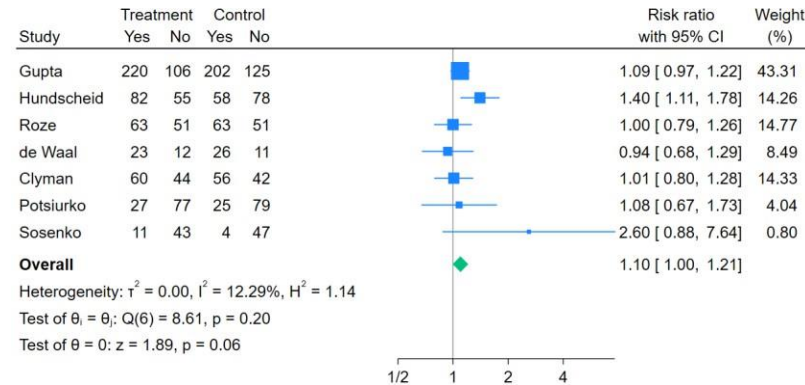
*Papers that provided median were converted to estimated mean and SD using the DECoMA tool (Chi KY, Li MY, Chen C, Kang E. Cochrane Taiwan. *Data Estimation and Conversion for Meta-Analysis* [DECoMA]. Cochrane Taiwan; 2022. Accessed September 8, 2024. <https://taiwan.cochrane.org/resource/decoma>).

eFigure 1. Risk of Bias Assessment



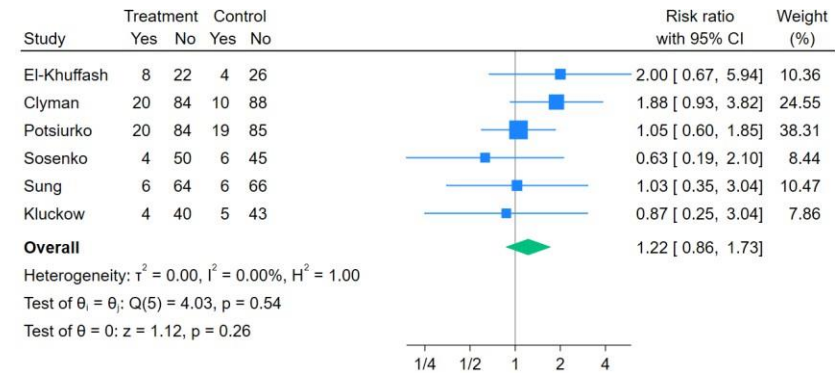
eFigure 2. Forest Plots for Primary and Secondary Outcomes Not Demonstrated in the Main Article

Death at 36 weeks or moderate/severe BPD at 36 weeks



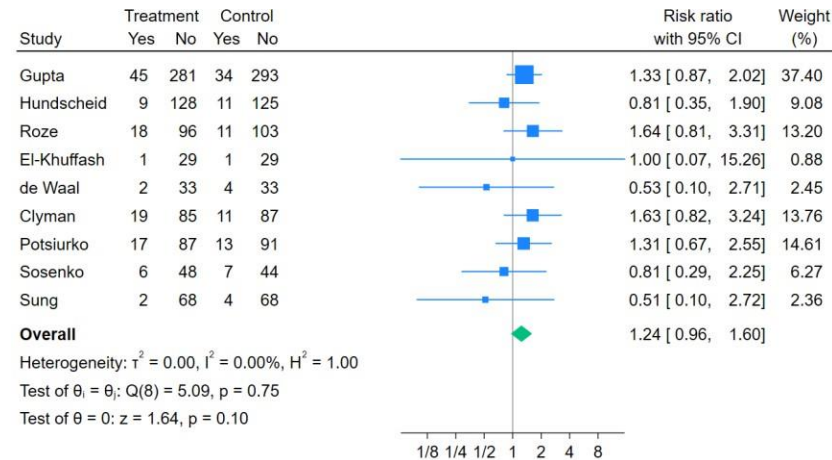
Random-effects REML model

Death before discharge



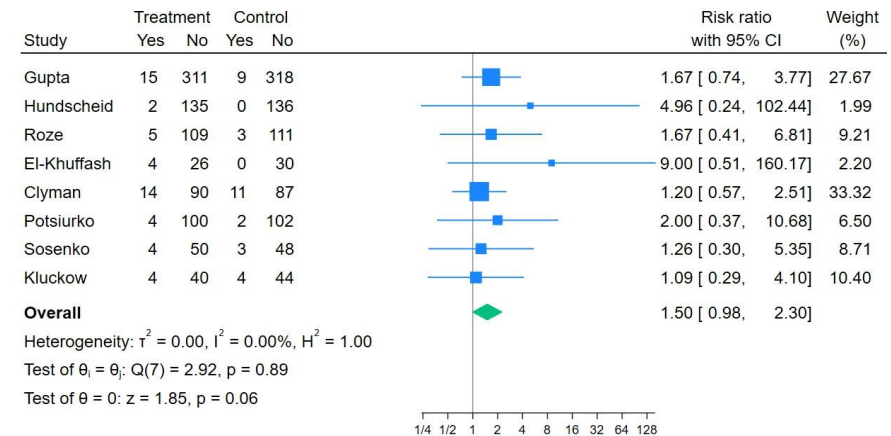
Random-effects REML model

IVH (grade III/IV with ventricular dilation) or intraparenchymal abnormality



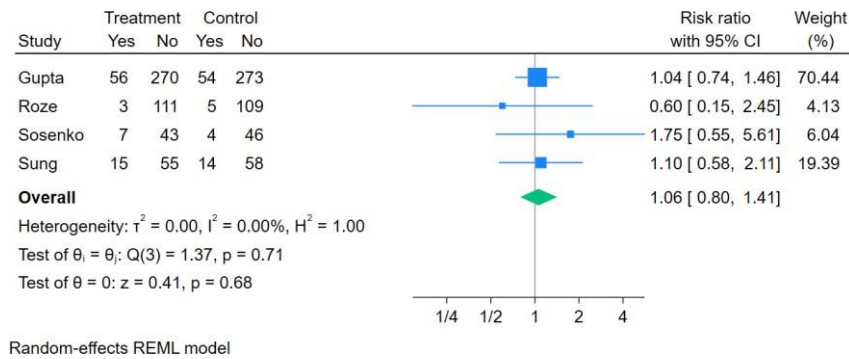
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PVL

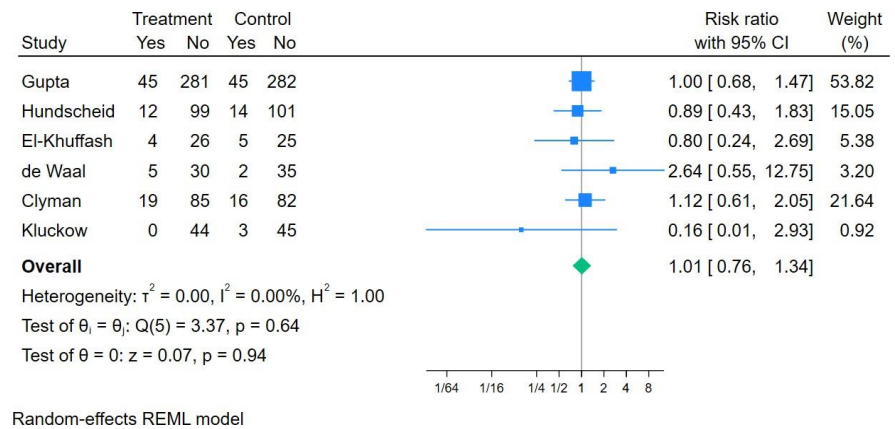


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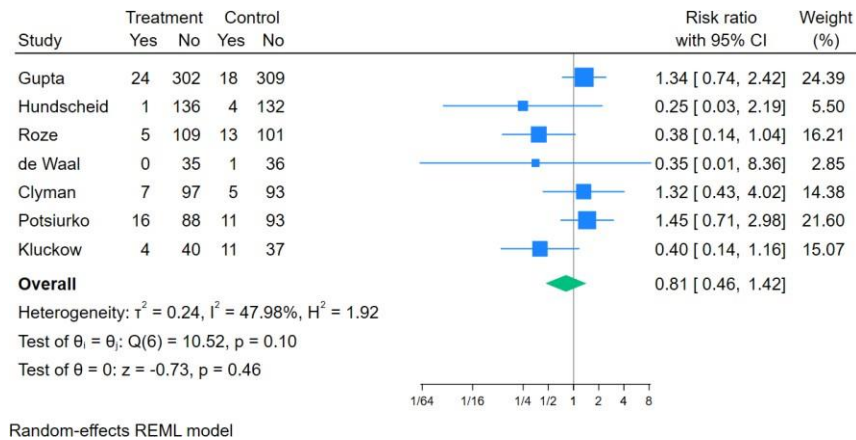
ROP (stage > 2)



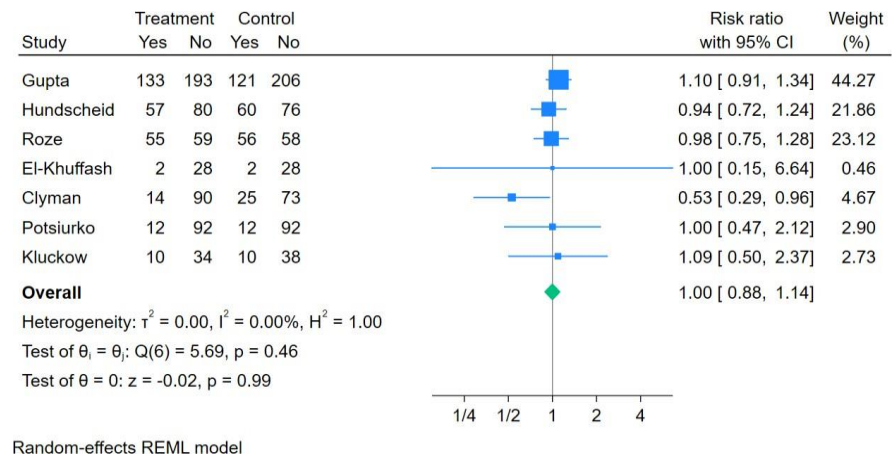
ROP treatment



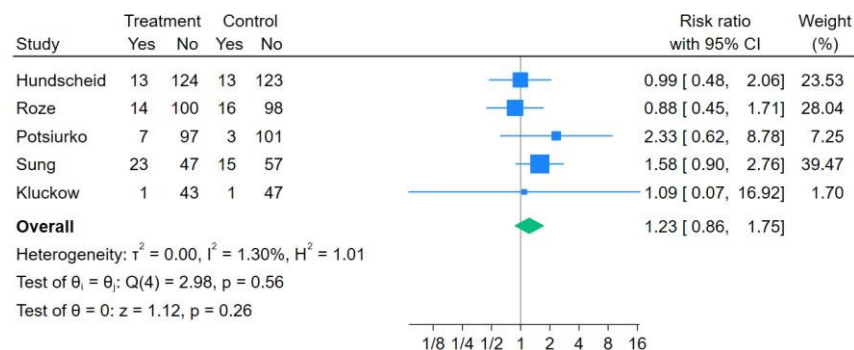
Pulmonary Hemorrhage



Hypotension OR inotrope/vasopressor OR fluid expansion

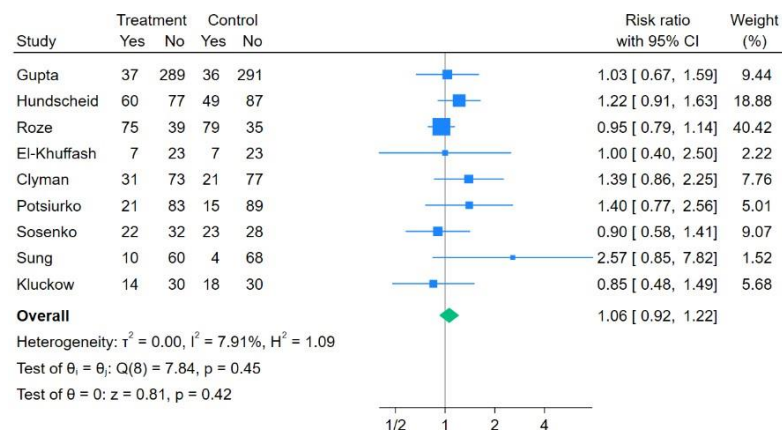


Kidney Failure



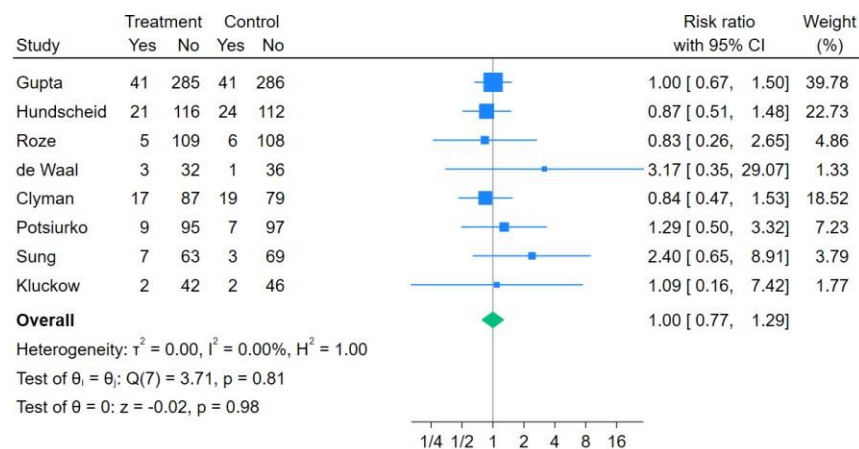
Random-effects REML model

Blood culture positive for sepsis



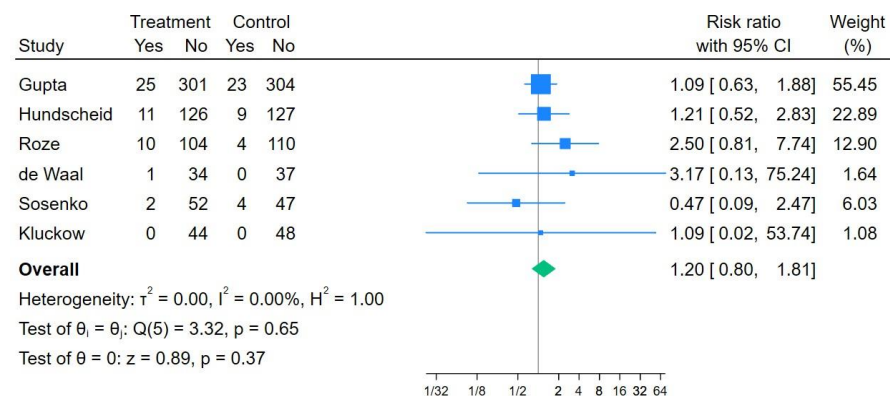
Random-effects REML model

NEC (stage $\geq$ II)

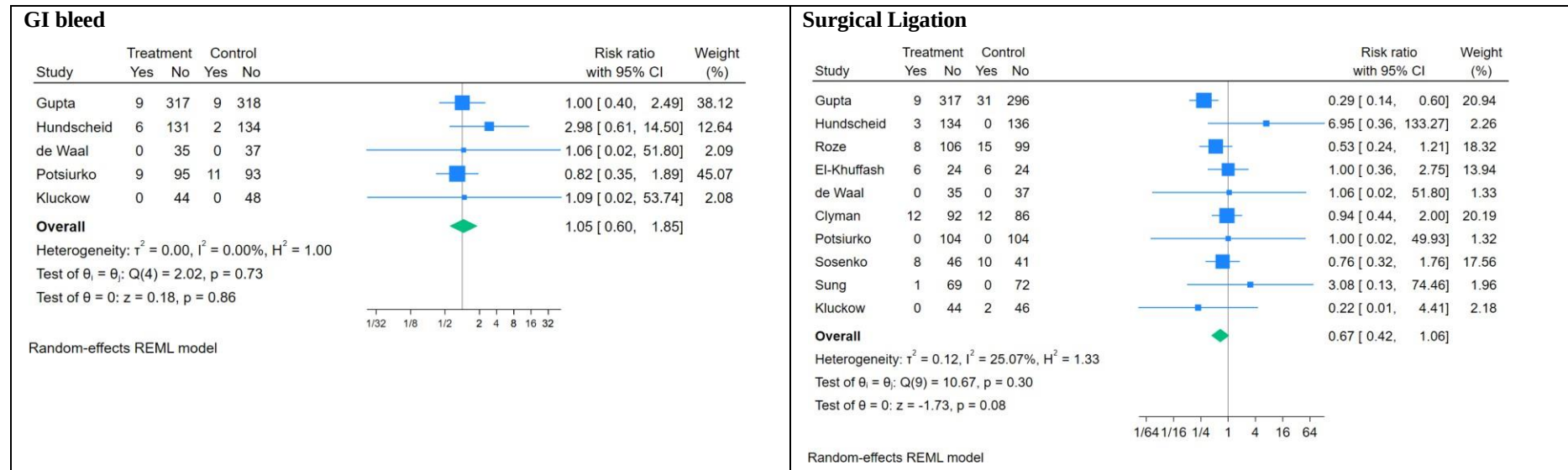


Random-effects REML model

GI perforation

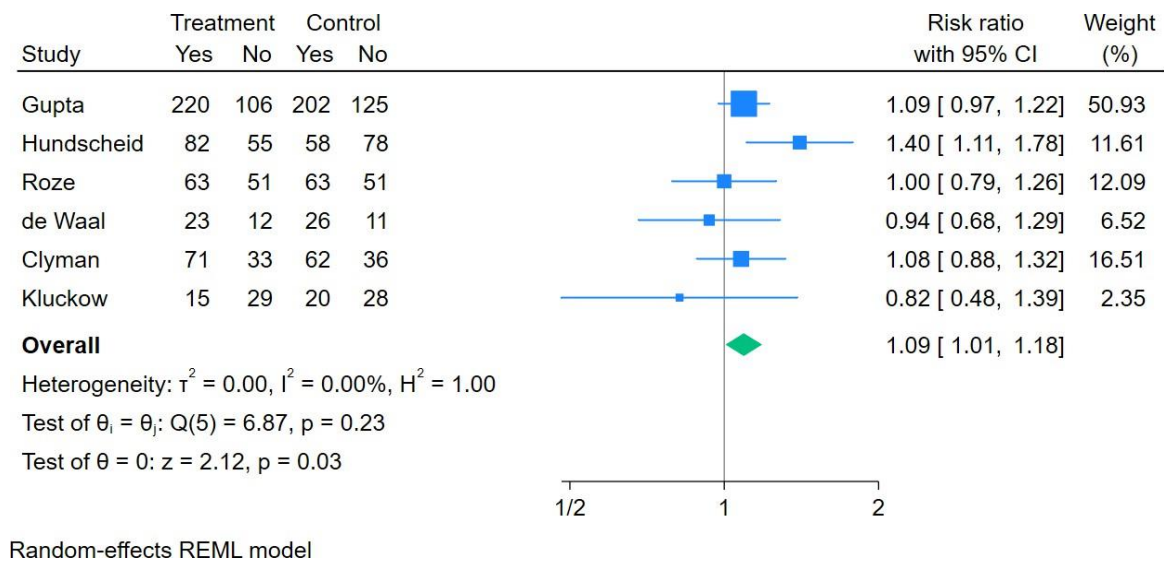


Random-effects REML model

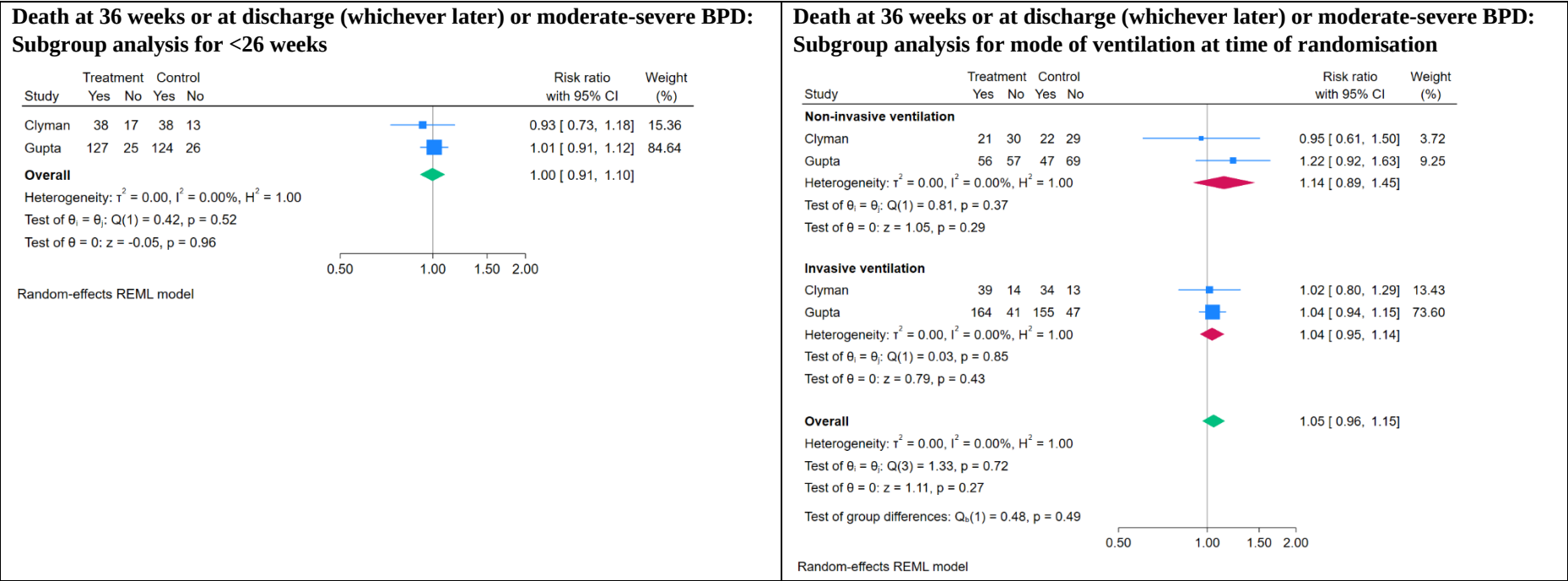


Abbreviations: BPD, bronchopulmonary dysplasia; GI, gastrointestinal; IVH, intraventricular hemorrhage; NEC, necrotizing enterocolitis; PVL, periventricular leukomalacia; ROP, retinopathy of prematurity.

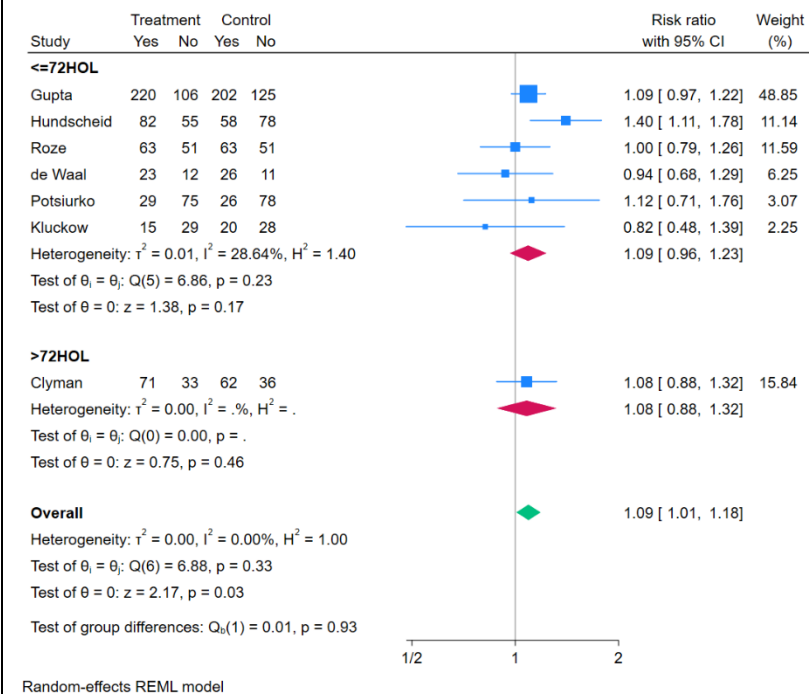
eFigure 3. Forest Plot Showing Subgroup Analysis at Less Than 29 Weeks for Composite Primary Outcome: Death at 36 Weeks or at Discharge (Whichever Occurred Later) or Moderate to Severe Bronchopulmonary Dysplasia in Figure 2A



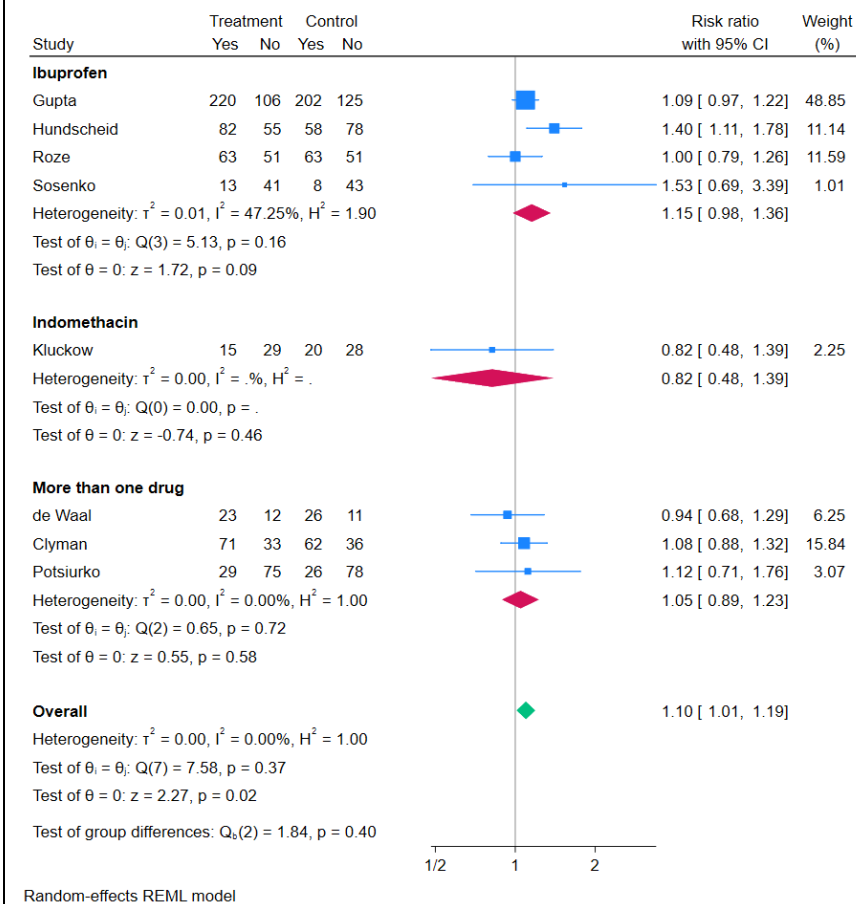
eFigure 4. Forest Plots of Additional Subgroup Analyses



Death at 36 weeks or at discharge (whichever later) or moderate-severe BPD: Subgroup analysis by hours of life

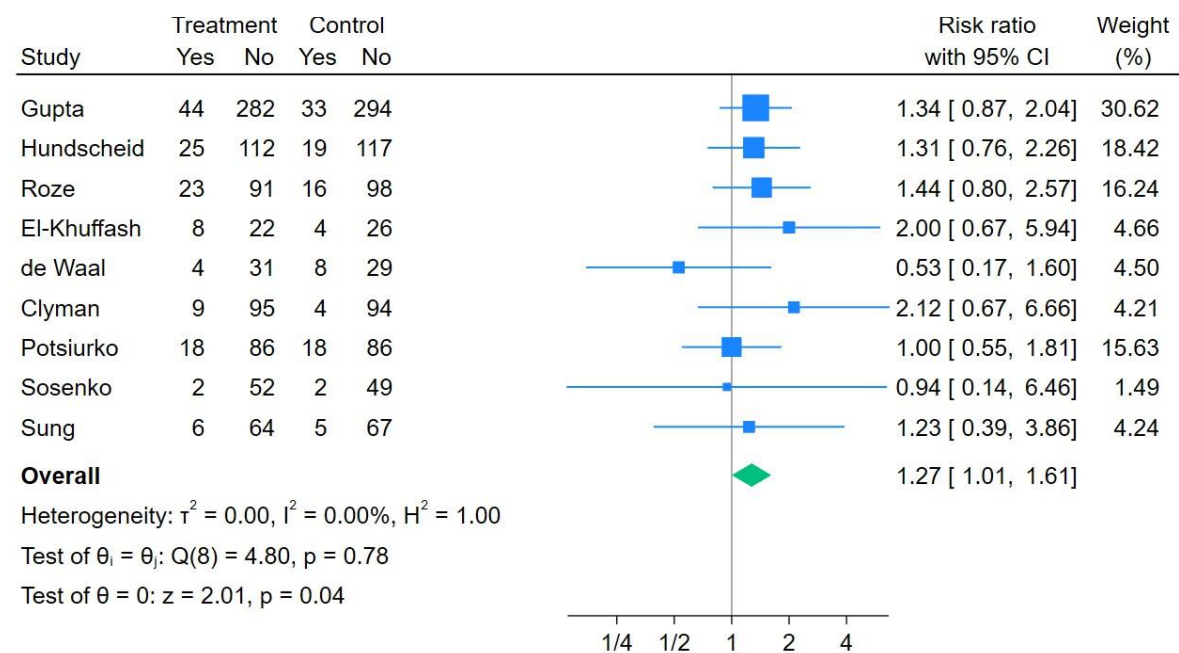


Death at 36 weeks or at discharge (whichever later) or moderate-severe BPD: Subgroup analysis by drug



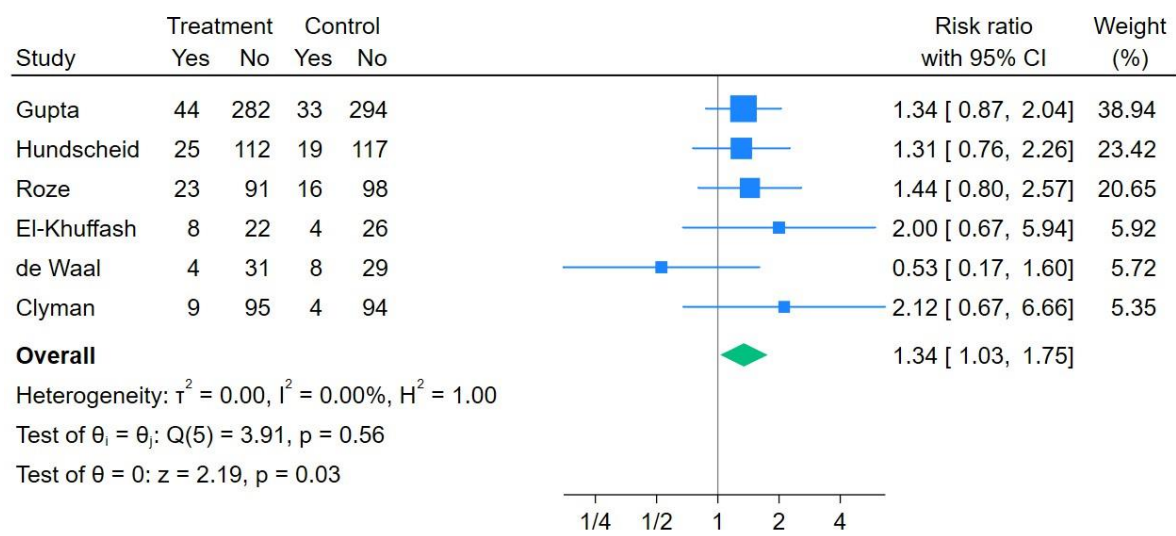
Abbreviations: BPD, bronchopulmonary dysplasia; HOL, hours of life.

eFigure 5. Forest Plot Showing Death at 36 Weeks



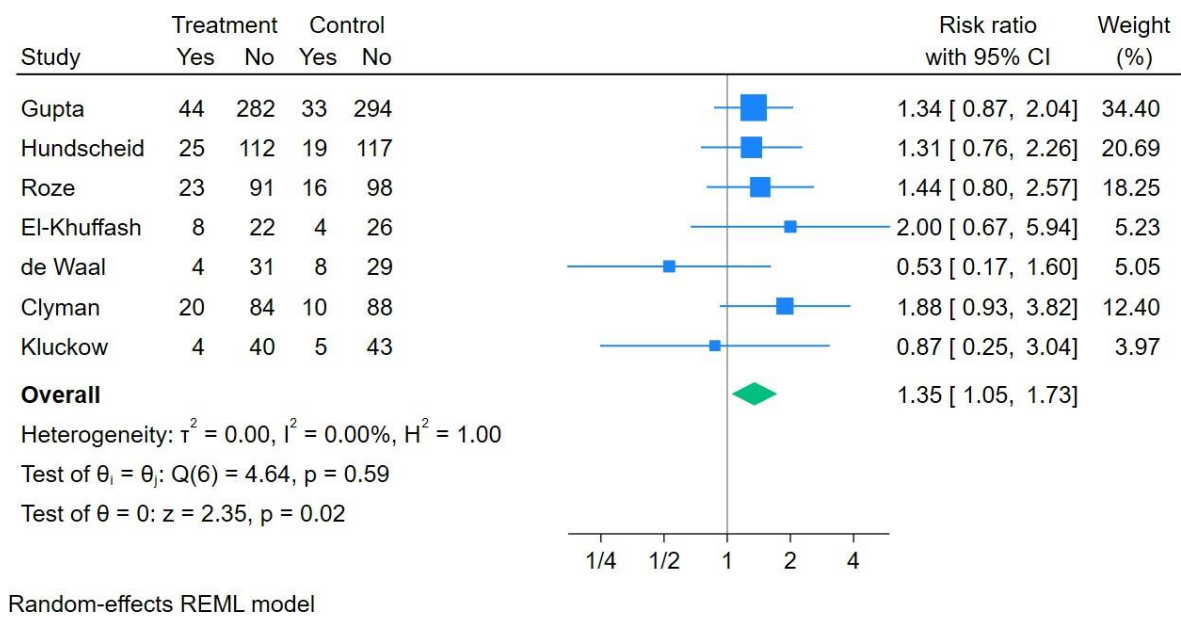
Random-effects REML model

eFigure 6. Forest Plot Showing Subgroup Analysis at Less Than 29 Weeks for Outcome: Death at 36 Weeks in eFigure 5

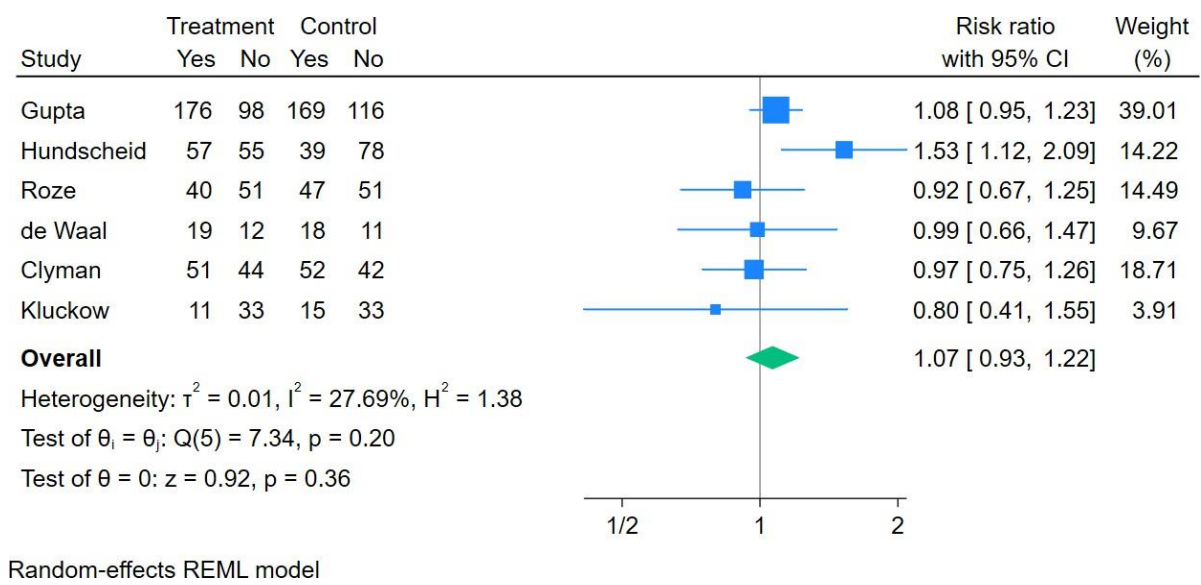


Random-effects REML model

eFigure 7. Forest Plot Showing Subgroup Analysis at Less Than 29 Weeks for Outcome: Death at 36 Weeks or at Discharge (Whichever Occurred Later) in Figure 2B



eFigure 8. Forest Plot Showing Subgroup Analysis at Less Than 29 Weeks: Moderate to Severe Bronchopulmonary Dysplasia at 36 Weeks in Figure 2C



eTable 7. Cause of Death

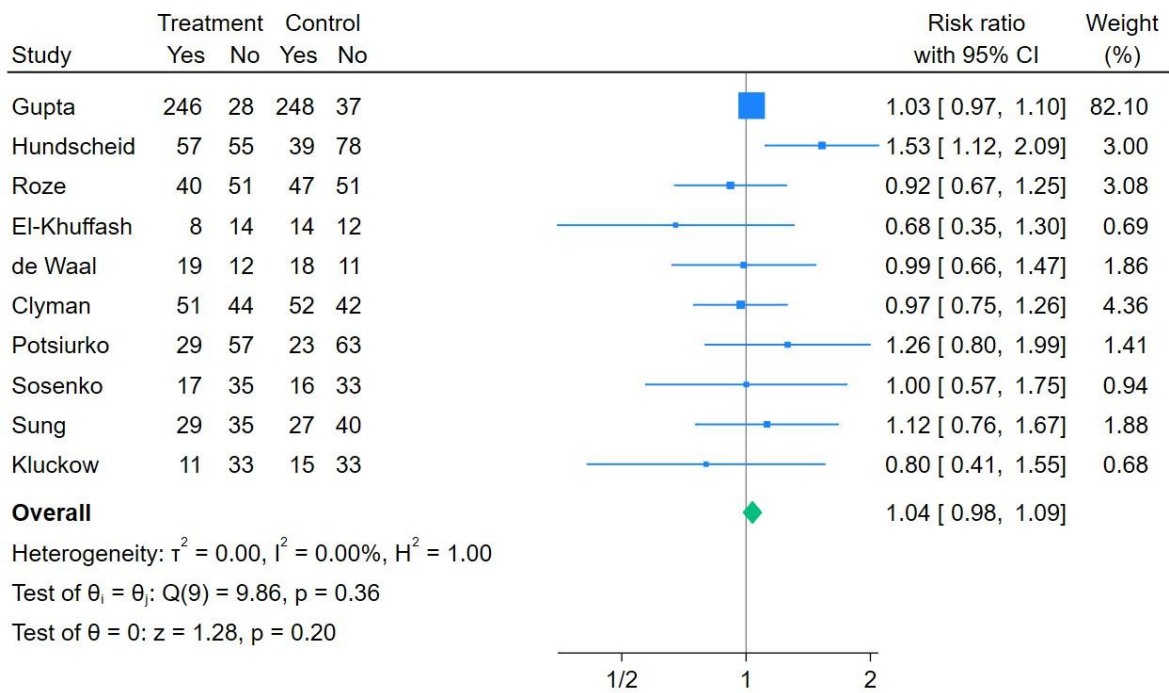
S/N	Study, Year		Treatment Arm	Expectant Arm
1	Sosenko et al., 2012 ²⁵	Not available		
2	Kluckow et al., 2014 ²⁶	Not available		
3	Clyman, 2019 ²⁷	Total deaths reported, n/n (%)	20/104 (19.2)	10/98 (10.2)
		BPD, n	0	1
		Intestinal obstruction/volvulus, n	1	1
		NEC, n	9	6
		Bacteremia (non-CoNS)	10	2
4	Hundscheid et al., 2023 ²⁸	Total deaths reported, n/n (%)	25/137 (18.2)	19/136 (14.0)
		Circulatory failure, n	4	1
		Respiratory insufficiency, n	2	2
		NEC/SIP, n	9	7
		Sepsis, n	9	7
		CNS injury, n	1	2
5	Gupta et al., 2024 ²⁹	Not available		
6	Rozé et al., 2021 ³⁰	Not available		
7	El-Khuffash et al., 2021 ³¹	Total deaths reported, n/n (%)	8/30 (26.7)	4/30 (13.3)
		Respiratory failure/pleural effusion, n	2	2
		NEC, n	3	1
		Sepsis, n	3	0
		IVH, n	0	1
8	de Waal et al., 2021 ³²	Total deaths reported, n/n (%)	4/35 (11.4)	8/37 (21.6)
		Respiratory failure	1	2
		IVH, n	1	-
		Sepsis, n	2	3
		NEC, n	-	2
		Kidney and liver failure	-	1
9	Potsiurko et al., 2024 ³³	Total deaths reported, n/n (%)	20/104 (19.2)	19/104 (18.3)
		NEC, n	6	3
		CNS, n	11	12
		Severe RDS, n	2	1
		Late-onset sepsis, n	1	3
10	Sung et al., 2020 ³⁴	Not available		
		Total	77	60

Abbreviations: BPD, bronchopulmonary dysplasia; CNS, central nervous system; CoNS, coagulase-negative staphylococci; IVH, intraventricular hemorrhage; NEC, necrotizing enterocolitis; RDS, respiratory distress syndrome; SIP, spontaneous intestinal perforation.

eTable 8. Cause of Death (Grouped by System)

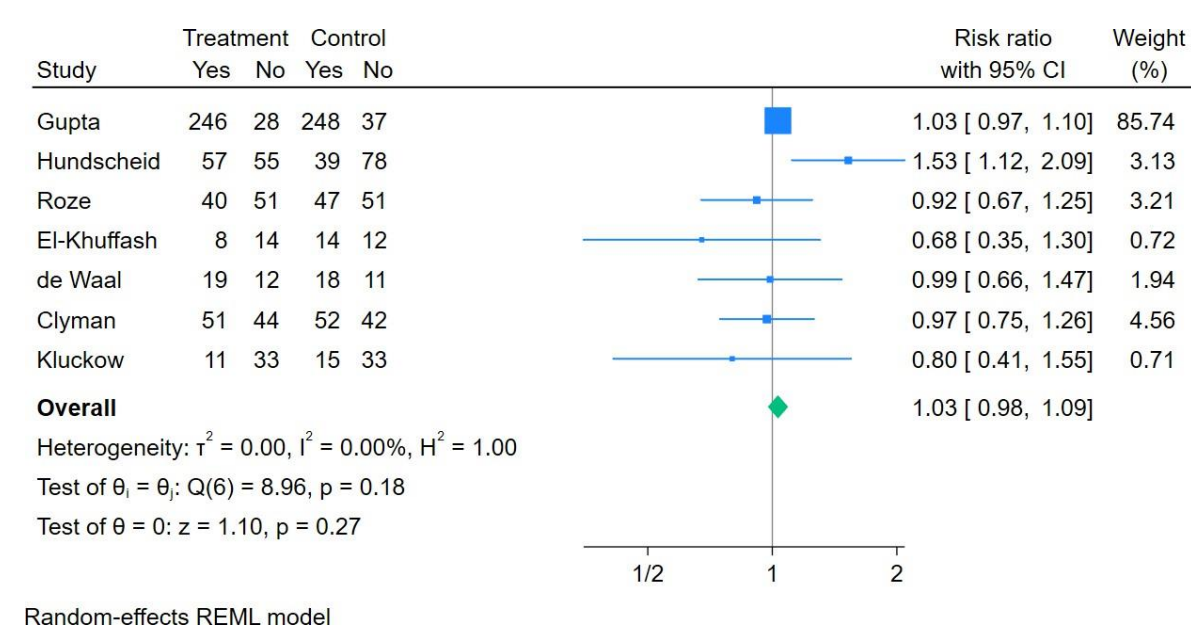
Cause of death	Treatment Arm	Expectant Arm
Central Nervous System	13	15
Respiratory	11	9
Gastrointestinal	28	20
Sepsis	25	15
Others	0	1
<i>Total</i>	<i>77</i>	<i>60</i>

eFigure 9. Forest Plot Showing Bronchopulmonary Dysplasia at 36 Weeks



Random-effects REML model

eFigure 10. Forest Plot showing Subgroup Analysis at Less Than 29 Weeks for Outcome: Bronchopulmonary Dysplasia (BPD) at 36 Weeks in eFigure 9



eTable 9. Secondary Outcomes for Which Meta-Analysis Was Not Done

Secondary Outcome	Result
Cause of Death	Five papers ^{27,28,31-33} (n=815) reported on cause of death from respiratory insufficiency, sepsis, or necrotising enterocolitis. Four papers (n=613) ^{28,31-33} reported on death due to central nervous injury. There were no statistically significant differences between the two arms. The cumulative cause of death by category is reported in eTable 8, and by individual paper in eTable 7.
Death at 28 days	One paper ²⁸ (n=273) reported on this outcome. The risk of death at 28 days was increased in the active treatment arm [RR (95% CI): 1.9 (1.02 – 3.57)].
Diuretics	Five papers ^{27-29,31,32} (n=1260) reported on this outcome. There was less use of diuretics in the active treatment arm compared to the expectant management arm (34.7% vs. 43.3%), which was statistically significant [RR (95% CI): 0.79 (0.69 – 0.91)].
Discharge home on oxygen support	One paper ²⁹ (n=653) reported on this outcome. 39.9% in the active treatment arm vs. 37.6% in the expectant management arm were discharged home on oxygen support. This was not statistically significant [RR (95% CI): 1.06 (0.87 – 1.29)].
Status of PDA at discharge	Two papers ^{25,33} (n=350) reported that 85.0% in the active treatment arm vs. 79.5% in the expectant management arm had closed PDAs prior to discharge. This was not statistically significant [RR (95% CI): 1.07 (0.97 – 1.18)]. Three papers ^{30,32,33} (n=508) reported on the PDA closure at day 10-14 of life. 166/253 (65.6%) in the active management arm vs 111/255 (43.5%) in the expectant management arm achieved this outcome. This was statistically significant [RR (95% CI): 1.51 (1.28 – 1.78)].
Weight gain	One paper ²⁹ (n=653) reported no mean difference in the Z scores of weight at birth and at discharge between both arms (mean difference –1 vs. –1.1; standard deviation of 1).
Duration of respiratory support	Six papers ^{25,28,30,31,33,34} report on duration of oxygen support (medians). Seven papers ^{25,28-32,34} report on days of invasive respiratory support (medians). Four papers ^{28-30,34} report on days of non- invasive respiratory support (medians).
Time to full enteral feeding	Five papers ^{26-28,33,34} report on time to full enteral feeding (medians and one mean).
Duration of hospital stay	Five papers ^{27,29,31-33} report on duration of hospital stay (medians).

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